

Project Design Phase Proposed Solution Template

Date	05 July 2026
Team ID	
Project Name	EV Analytics & Visualization Dashboard
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	City planners, fleet managers, and everyday EV buyers all struggle to make sense of EV data. Planners can't easily see charging demand patterns, fleet managers can't tell how battery health is affecting real driving range, and consumers have no easy way to compare EV models on more than price. The data exists, but it's scattered and not visualized in one place.
2.	Idea / Solution description	A Tableau-based EV Analytics & Visualization Dashboard that ingests charging station, vehicle sensor, and EV model data, cleans and stores it in a central warehouse, and presents it through three tailored dashboards: charging pattern & usage analysis, battery performance vs. driving range, and comparative EV model efficiency.
3.	Novelty / Uniqueness	Brings three very different stakeholders (city planner, fleet manager, consumer) onto one platform with role-specific views, directly correlates battery health metrics with real-world driving range instead of just showing raw specs, and lets consumers filter model comparisons by their own driving style and region for a personalized recommendation.
4.	Social Impact / Customer Satisfaction	Helps cities plan charging infrastructure and manage grid load more sustainably, reduces fleet downtime and maintenance costs through proactive battery insights, and gives consumers the confidence to choose the right EV — all of which support faster, smoother EV adoption and lower emissions.
5.	Business Model (Revenue Model)	SaaS subscriptions for city/government agencies and fleet operators (tiered by number of vehicles or seats); a freemium consumer-facing comparison tool with premium features (personalized reports, alerts) behind a paywall; and data-partnership or licensing revenue from charging networks and EV manufacturers.
6.	Scalability of the Solution	The cloud-hosted data pipeline and warehouse scale horizontally as more vehicles, charging stations, and users come online. The modular architecture allows new data sources (additional EV brands, new cities) and new dashboard types to be added without redesigning the core system.